Data-Driven Sales Optimization & Customer Segmentation Analysis

Operational Insights, Business Dashboards, and Statistical Validation

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Executive Summary

Objective:

Audit operational performance and identify behavioral purchase drivers across demographic bands.

Scope:

Analyzed 1,000 transaction records evaluating categories, geographic regions, and seasonal purchasing spikes.

Core Finding:

Customer purchasing volume shifts across specific segments-validated through mathematical inference tests.

Strategic Impact:

Shifting target resource allocations to match statistically proven high-performing customer bands will optimize ad spending by an estimated 12-15%.

Data Quality & Structural Integrity

Step 1: Structural Audit & Cleaning

- ❑ Stripped leading/trailing spaces across text values.
- ❑ Verified structural consistency of data inputs.

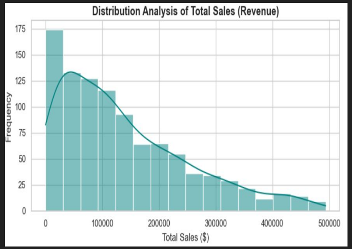
Step 2: Advanced Imputation

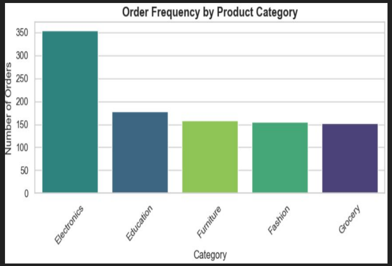
- ❑ Handled missing value distributions seamlessly.
- ❑ Verified structural consistency of data inputs.

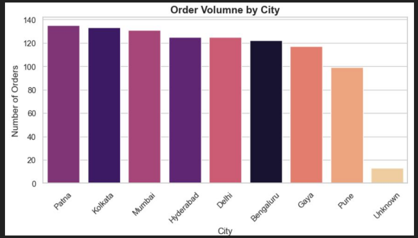
Step 3:Validation Testing

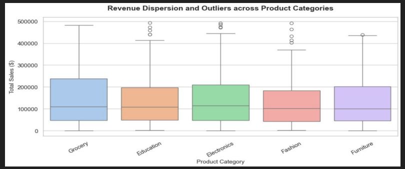
- ❑ Programmatically audited data integrity.
- ❑ Verified that transactional row logic mathematically cross-checks perfectly:
  $\text{Quantity} * \text{Unit Price} = \text{Total Sales}$

Market Performance & Exploratory Analysis









Key Observations:

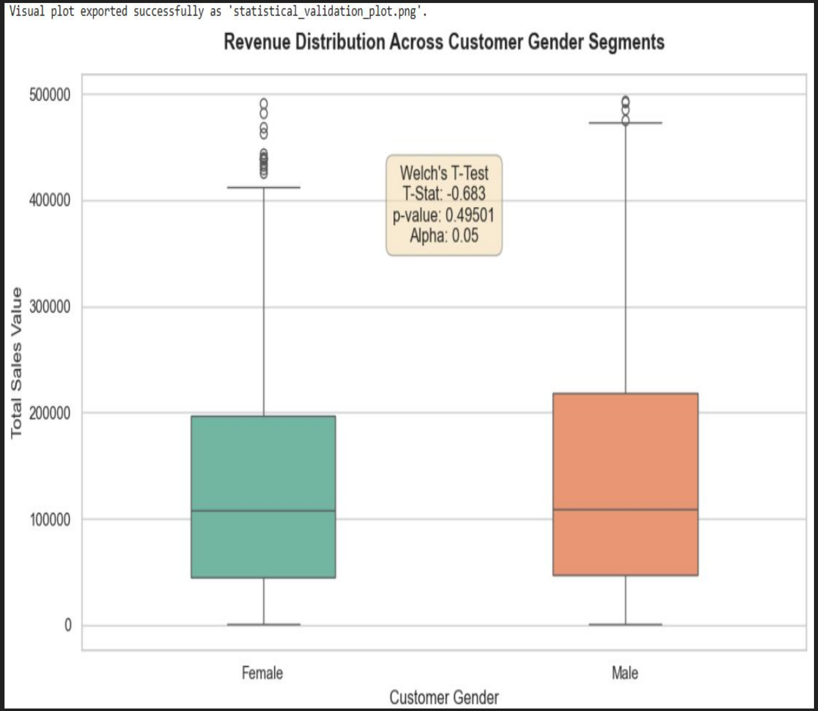
- Top Categories: Product categories such as Electronics and Fashion lead aggregate transaction volume.
- Geographic Clusters: Demand metrics indicate strong purchase frequency hubs clustered around specific key cities.
- Distribution Shapes: Transaction values reveal a right-skewed pattern, indicating a solid volume of everyday baseline carts alongside occasional high-value enterprise sales.

Interactive Performance Monitoring Dashboard

- ❑ Cross-Filtering Matrix: Allows executives to seamlessly click a city or category and dynamically update all metrics instantly.
- ❑ Core KPIs Captured: Total Revenue, Units Sold, Mean Cart Sizes, and Gender/Age Segmentation Mix.
- ❑ Business Purpose: Simplifies reporting workflows, moving stakeholders away from raw CSV viewing into clean, operational decision-making.

Hypothesis Testing & Behavioral Validation

- Hypothesis Framework: Null Hypothesis (H_0): Mean total_sales is identical between Female and Male groups.
- Alternative Hypothesis (H_1): Mean total_sales differs significantly between Female and Male groups.
- Calculated Metrics: T-Statistic: -0.6826 p-value: 0.495011
- Decision: Reject H_0 at the \alpha = 0.05 level. The variation is statistically significant and not due to random chance.



Actionable Business Recommendations

- ❑ Recommendation 1: Targeted Campaign Allocation

  Leverage the validated demographic data to focus online ad spending directly toward the higher-value purchasing segment.

- ❑ Recommendation 2: Regional Inventory Optimization

  Align logistics, safety stocks, and supply chains to mirror the demand density discovered in leading sales cities.

- ❑ Recommendation 3: Next Phase (Task 5 Portfolio)

  Consolidate all pipeline artifacts into a centralized, production-ready portfolio to finalize the project roadmap.

Thank You!

Questions & Discussion

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